EIGHT INDUSTRY AI PRODUCT CASE STUDY WORKBOOK

Management of AI Products | EC2 Closed-Book Preparation

Narrative cases + direct topic links + three examination questions + red model answers

Design principle

Each case is written as a real business situation. Topic names appear only after the case, so you first learn to recognise topics from business facts. Every case then has three connected sub-questions: A for framing, B for product design, and C for evaluation and recommendation.

Important

Do not memorise the company or industry wording. Memorise how the product manager moves from problem to people, Artificial Intelligence role, data, Minimum Viable Product, uncertainty, human control, testing, measures, monitoring, and final decision.

Eight cases: Healthcare | Banking | Manufacturing | Retail | Public Services | Education | Logistics | Enterprise Software

How to Use This Workbook

For each industry, read the case story without looking at the topic links. Write the likely business problem, stakeholders, Artificial Intelligence output, data, risks, and product decision on rough paper. Then compare with Direct Topic Links. Finally, attempt Questions A, B, and C before reading the red model answers.

Question	Likely thinking level	What to prepare
A	Frame and identify	Role, problem, stakeholders, Artificial Intelligence usefulness, technique, data
B	Apply and design	Minimum Viable Product, workflow, user experience, uncertainty, Human-in-the-Loop
C	Evaluate and recommend	Errors, testing, measures, deployment, monitoring, rollback, final decision

Universal answer opening

As the product manager in this case, I will...

CASE 1: CareFirst Hospital Appointment and Triage Assistant

Healthcare

Why this case matters

Healthcare tests high-consequence errors, data sensitivity, uncertainty, professional judgment, and safe Human-in-the-Loop design.

Business Case

CareFirst Hospital receives about 7,000 patient requests every week through telephone calls, a website, and walk-in counters. Requests include appointment booking, rescheduling, payments, report status, and free-text descriptions of symptoms in several languages. The average call waiting time is thirteen minutes, and 18 per cent of appointments are manually transferred because patients selected the wrong department.

Hospital management asks the Artificial Intelligence product manager to explore an assistant. The requested product should understand speech or text, identify request type, suggest the correct department, and highlight requests that may need urgent clinical review. Doctors are concerned that the product may appear to diagnose illness. Data-governance staff are concerned about sensitive patient information and unequal performance across languages.

A pilot model identifies 92 of 100 urgent requests but also marks 160 of 1,000 non-urgent requests as urgent. Clinical staff can review only 120 additional requests per day. The product manager must decide the first product scope, how the interface should communicate uncertainty, and whether the model is ready for real operation.

Direct Topic Links

Topic	Relevance in this case
Opportunity framing	Reduce waiting and routing errors; do not begin with a chatbot.
Stakeholders	Hospital buyer, patients, appointment staff, clinical triage, doctors, data-governance team.
Artificial Intelligence technique	Speech-to-text, Natural Language Processing, classification, possible regression for waiting time.
Data	Patient request, language, transcript, verified route, clinical review outcome, appointment system.
Minimum Viable Product	Administrative routing plus supervised clinical escalation; no autonomous diagnosis.
Uncertainty	Separate confirmed hospital information from urgency inference and clinical recommendation.
Human-in-the-Loop	Appointment staff resolve administrative ambiguity; qualified clinical staff review symptom and urgency cases.
Errors and metrics	False negative may delay care; false positive may overload clinical reviewers.
Testing	Shadow deployment, limited beta, language-level guardrails, rollback.

How to Read the Case

- High-consequence domain
- Free text and speech
- Classification with serious false negatives
- Sensitive data
- Qualified professional review

Product Development Link

Patient request -> language and transcript verification -> fixed administrative retrieval -> request classification -> urgency inference -> confidence and risk check -> administrative completion or qualified human escalation -> verified outcome -> monitoring and improvement.

Professor-Style Questions and Answers

Question A, 5 marks: Frame the healthcare Artificial Intelligence opportunity

Identify the real business problem, stakeholders, suitable Artificial Intelligence role, data required, and parts that should remain rule based.

MODEL ANSWER

As the product manager, I will build an appointment-routing and urgency-support product, not an autonomous diagnosis product.

- The buyer is CareFirst Hospital; direct operational users are appointment and clinical triage staff; patients are service users and beneficiaries.
- Natural Language Processing will identify language and request intent. Classification will route requests and estimate an urgency band.
- Confirmed appointment slots, fees, hours, and published policies remain deterministic retrieval and rules.
- Required data includes verified transcripts, request type, correct department, clinical review outcome, language, and appointment result.
- Sensitive data, language representation, transcription errors, and historical routing bias require controls.

Recommendation: Begin with a narrow administrative and supervised-triage opportunity, keeping diagnosis and treatment outside product authority.

Question B, 5 marks: Design the product experience and human control

Design the Minimum Viable Product, workflow, facts, inferences, recommendations, and Human-in-the-Loop.

MODEL ANSWER

The Minimum Viable Product will cover one outpatient department, two languages, appointment administration, department routing, and escalation of symptom-related requests.

- Confirmed appointment availability and hospital policy are facts with source and update time.
- “Likely cardiology request” or “possible urgent review” is an inference and must be labelled as an estimate.
- “Route to clinical triage within fifteen minutes” is a recommendation for staff, not a diagnosis for the patient.
- Low-confidence, conflicting, symptom-related, and unsupported-language cases go to qualified staff.
- Patients can correct transcript and details; staff can override routing and record the reason.

Recommendation: Keep medical judgment with qualified staff and automate only low-risk administrative work.

Question C, 5 marks: Evaluate readiness and deployment

Evaluate the pilot errors and recommend measures, testing, capacity controls, and rollout.

MODEL ANSWER

The model catches 92 per cent of urgent cases but misses eight urgent requests and creates 160 false alarms among 1,000 non-urgent requests.

- Missed urgent requests are false negatives and may delay care.
- False urgent alerts may exceed clinical capacity and delay truly urgent patients.
- Measure urgent-case recall, precision, false-alert volume, performance by language, calibration, correct routing, waiting time, review workload, and complaints.
- Run historical testing and shadow deployment before a selected-user beta.
- Expansion requires agreed safety, workload, and language guardrails with manual fallback and rollback.

Recommendation: Do not approve full deployment using one headline accuracy figure; require safe workflow performance and manageable review demand.

Case Memory Box

Remember

Healthcare = high consequence. Use Artificial Intelligence for routing and risk support, but separate facts from clinical inference and retain qualified human judgment.

CASE 2: SafePay Fraud Detection and Customer Protection

Banking and Financial Services

Why this case matters

Banking tests classification, anomaly detection, false positives, false negatives, investigator capacity, customer communication, audit, and Agentic Artificial Intelligence authority.

Business Case

SafePay Bank processes two million card transactions every day. Existing fraud rules block 12,000 transactions daily, but investigators confirm only 900 as fraud. Customers complain that legitimate purchases are declined, and merchants complain about lost sales.

A new Machine Learning model finds 1,050 fraudulent transactions but blocks 15,000 transactions. Investigators can review only 4,000 cases per day. The product team wants an autonomous agent to block cards, message customers, initiate refunds, and close cases. Compliance staff require explanations and an audit trail.

The product manager must decide how rules and Machine Learning should work together, which actions may be automated, how customers can correct a decision, and whether the new model creates enough value to justify additional friction and review workload.

Direct Topic Links

Topic	Relevance in this case
Business problem	Reduce fraud loss without unnecessary declines and customer harm.
Technique	Classification, anomaly detection, risk scoring, ranking.
Rules versus Machine Learning	Known policy conditions remain rules; variable risk patterns use Machine Learning.
Errors	False positive blocks genuine payment; false negative allows fraud.
Capacity	Alerts must fit investigator and customer-verification capacity.
Uncertainty and explainability	Show transaction fact separately from fraud suspicion.
Agentic authority	Reversible verification may be automatic; high-impact blocks and refunds need limits or approval.
Metrics	Fraud value prevented, legitimate value blocked, precision, recall, review load, complaints, conversion.
Monitoring	Fraud patterns change; monitor drift, provider change, complaints, and rollback.

How to Read the Case

- Rare positive event
- Customer harm from false alarms
- Limited investigators

- Autonomous financial action
- Audit requirement

Product Development Link

Transaction facts -> rules and risk model -> risk tier -> approve, verify, review, or temporary hold -> customer correction or investigator decision -> confirmed fraud outcome -> monitoring and retraining.

Professor-Style Questions and Answers

Question A, 5 marks: Compare the rules and new model

Should SafePay replace the current rules with the new model? Use errors, customer impact, and capacity.

MODEL ANSWER

I will not replace the complete rule system immediately because the new model detects 150 more fraud cases but blocks 3,000 additional transactions.

- False positives create genuine declines, complaints, support cost, lost sales, and possible customer loss.
- False negatives permit fraud loss and customer harm.
- Investigator capacity is far below total blocked volume, so cases need risk tiers and ranking.
- Use strong rules for known patterns and the model for variable risk scoring and prioritisation.
- Compare fraud value prevented, legitimate value blocked, precision, recall, review capacity, and customer correction.

Recommendation: Run a controlled economic and customer-impact pilot instead of selecting the model using detected fraud count alone.

Question B, 5 marks: Design authority and customer experience

Define the autonomous agent's authority, Human-in-the-Loop, customer communication, and fallback.

MODEL ANSWER

I will divide action authority by risk, reversibility, evidence, and value.

- Low-risk action: send secure verification or request step-up authentication.
- Medium-risk action: temporary reversible hold after multiple strong signals with rapid customer confirmation.
- High-risk action: permanent block, large refund, or account restriction requires investigator approval and policy checks.
- The message states the transaction as fact and suspicion as inference; it provides a secure correction route without revealing exploitable rules.
- All actions and overrides are logged, and current rules plus human support remain fallback.

Recommendation: Move from recommendation to limited reversible action only after evidence and control testing.

Question C, 5 marks: Test and monitor the product

Recommend a Minimum Viable Product, deployment approach, and success measures.

MODEL ANSWER

The Minimum Viable Product will use the new model for risk ranking and customer verification on selected transaction types without autonomous permanent blocking.

- Run offline evaluation, shadow scoring, and a limited beta on defined merchants or transaction categories.
- Measure precision, recall, fraud value, genuine value declined, investigator workload, customer response, complaint rate, and conversion.
- Check results by customer, merchant, geography, and transaction type.

- Monitor changing fraud patterns, data quality, external provider behaviour, and cost.
- Pause or roll back if genuine-customer harm or review load exceeds guardrails.

Recommendation: Expand only when additional fraud prevented exceeds customer, operational, and financial harm.

Case Memory Box

Remember

Banking = balance fraud loss against genuine-customer friction. Accuracy alone is not enough; authority, capacity, explanation, correction, and audit are central.

CASE 3: Axis Manufacturing Predictive Maintenance

Manufacturing and Industrial Operations

Why this case matters

Manufacturing tests sensor data, anomaly detection, classification, regression, operational capacity, economic consequence, shadow deployment, feedback, drift, and rollback.

Business Case

Axis Manufacturing operates 240 machines across three plants. An unexpected breakdown causes an average production loss of Rs. 4.8 lakh per hour. The company records temperature, vibration, electrical load, maintenance notes, operating hours, and technician inspections.

A model predicts whether a machine will fail within seven days. During a three-month test, there were 40 actual failures. The model correctly warned about 34 failures, missed 6 failures, and created 52 false alarms. Each inspection costs Rs. 18,000, and the maintenance team can inspect only 20 machines per day.

Management wants automatic work orders. Technicians argue that sensor quality differs across plants and that some false alarms arise after maintenance or changes in operating load. The product manager must connect model output with real maintenance decisions and production value.

Direct Topic Links

Topic	Relevance in this case
Technique	Anomaly detection, classification, regression, ranking.
Data	Sensor trends, load, age, maintenance, operating hours, verified inspection outcome.
Data risk	Calibration differences, missing readings, inconsistent notes, leakage, drift.
Errors	Missed failure versus unnecessary inspection.
Operational capacity	Rank alerts because inspection resources are limited.
Minimum Viable Product	One plant and selected machine family, human-approved inspection.
Deployment	Shadow mode, selected beta, canary by machine family or plant.
Feedback	Verified technician finding becomes the outcome label.
Metrics	Avoided downtime, warning lead time, inspection cost, false alarm, missed failure, adoption.

How to Read the Case

- Sensor data
- Failure prediction
- High cost of missed event
- Expensive false alarm
- Limited technician capacity

Product Development Link

Sensor and maintenance data -> anomaly and failure-risk model -> ranked alert with evidence -> planner review -> technician inspection -> verified condition and action -> production outcome -> controlled feedback and model monitoring.

Professor-Style Questions and Answers

Question A, 5 marks: Translate the business problem

Identify the business decision, technique, data, labels, and important data risks.

MODEL ANSWER

The business decision is which machines require inspection or planned maintenance within the next seven days.

- Anomaly detection identifies unusual sensor patterns; classification predicts failure within seven days; regression may estimate remaining useful time.
- Features include trend, load, age, operating hours, machine type, and maintenance history.
- Labels must come from confirmed failures and verified inspection outcomes, not previous alerts.
- Risks include sensor calibration, missing data, inconsistent notes, changing load, and information collected after failure.
- Performance must be checked by plant, machine type, age, and operating condition.

Recommendation: Begin with the maintenance decision and dependable data pipeline, not the most complex model.

Question B, 5 marks: Evaluate the model and design the workflow

Is the model ready for operational use? Include false positives, false negatives, capacity, and human control.

MODEL ANSWER

The model is suitable for a limited, human-supervised pilot, but not autonomous maintenance action.

- Six missed failures are false negatives and may cause major production loss.
- Fifty-two false alarms could create Rs. 9.36 lakh in inspections if all are investigated.
- The model detected 34 of 40 failures, but alert timing and precision also matter.
- Because capacity is twenty inspections daily, alerts must be ranked by risk, consequence, confidence, and lead time.
- The planner sees evidence and approves inspection; technicians verify actual condition.

Recommendation: Use the model as ranked decision support and adjust thresholds using operational and economic evidence.

Question C, 5 marks: Deploy, monitor, and improve

Recommend the Minimum Viable Product, testing, measures, feedback, and rollback.

MODEL ANSWER

The Minimum Viable Product will cover one plant, one machine family, ranked alerts, evidence, planner approval, and technician outcome capture.

- Run shadow deployment and compare predictions with verified failures and inspections.
- Move to a limited beta only after alert quality and workload guardrails are met.
- Measure missed failures, false alarms, warning lead time, downtime avoided, inspection cost, technician acceptance, and safety.
- Only verified outcomes enter retraining; monitor sensor quality, drift, alert volume, and plant-level performance.
- If quality falls or workload exceeds capacity, pause automation and roll back to baseline scheduling.

Recommendation: Expand plant by plant only after repeatable real-world value is demonstrated.

Case Memory Box

Remember

Manufacturing = model output must fit maintenance capacity and economics. Verified field outcomes, shadow deployment, and drift monitoring are essential.

CASE 4: FreshBasket Grocery Recommendation and Demand Product

Retail and E-commerce

Why this case matters

Retail tests customer journey, recommendation, market-basket analysis, rules versus Machine Learning, experimentation, sentiment, demand forecasting, sensitive inference, and business measures.

Business Case

FreshBasket serves 400,000 monthly online grocery customers. Thirty per cent of baskets contain repeat purchases, while 25 per cent of customers leave search without adding an item. Availability differs by store and changes throughout the day.

The company wants personalised repeat-item suggestions, complementary products, and substitutes for unavailable products. A pilot raises add-to-basket rate from 18 per cent to 22 per cent, but unavailable products are recommended in 9 per cent of sessions. Some customers complain that the system appears to infer sensitive dietary or health conditions.

The merchandising team also wants to use demand forecasts for stock planning. The product manager must distinguish recommendation, inventory rules, demand forecasting, experimentation, and privacy while measuring completed customer value rather than clicks alone.

Direct Topic Links

Topic	Relevance in this case
Recommendation	Rank repeat, complementary, and substitute products.
Market-basket analysis	Find products frequently purchased together.
Rules	Inventory, legal restrictions, confirmed allergens, and opt-out.
Demand forecasting	Predict product, store, and time demand using regression or forecasting.
Customer journey	Place recommendations where they help discovery or substitution.
Experiment	A/B test completed useful baskets, not clicks alone.
Data and privacy	Use explicit preferences; avoid unjustified sensitive inference.
Metrics	Basket completion, availability, substitution acceptance, margin, retention, complaints.
Sourcing	Build or subscribe with data, quality, cost, and lock-in controls.

How to Read the Case

- Ranking and recommendations
- Customer journey and placement

- Changing inventory
- A/B testing
- Sensitive personalisation

Product Development Link

Customer context and inventory -> candidate products -> rules and ranking -> recommendation or substitute -> customer accepts, ignores, or corrects -> completed basket and margin outcome -> experiment and monitoring.

Professor-Style Questions and Answers

Question A, 5 marks: Map the opportunity and techniques

Explain which parts need recommendation, market-basket analysis, forecasting, or fixed rules. Identify the data and privacy concerns.

MODEL ANSWER

I will separate repeat recommendation, complementary recommendation, substitute ranking, inventory validation, and demand planning.

- Recommendation ranks repeat and complementary items using purchase and current-basket context.
- Market-basket analysis identifies items frequently bought together.
- Demand forecasting predicts product, store, and time demand for planning.
- Real-time availability, restrictions, confirmed allergens, and explicit opt-out remain deterministic rules.
- Use transaction, search, basket, inventory, substitution, season, and explicit preference data; avoid hidden sensitive inference without justified purpose and permission.

Recommendation: Use Machine Learning for ranking and forecasting but keep safety, availability, and customer-choice constraints rule based.

Question B, 5 marks: Design the customer experience and experiment

Design the product journey and A/B test to determine whether the product creates value.

MODEL ANSWER

Recommendations will appear where they reduce discovery effort or solve an unavailable-item problem, not simply wherever screen space exists.

- Show repeat items early, complementary suggestions after basket context exists, and substitutes only when availability changes.
- Explain why a substitute is suggested and allow customers to reject or correct preferences.
- Compare a control group with a similar test group by store, customer type, and device.
- Primary product outcome is completed useful basket or accepted substitution, not click rate alone.
- Guardrails include unavailable recommendations, opt-out, complaints, sensitive inference, and segment-level performance.

Recommendation: Expand only when completed customer value, margin, and retention improve without intrusive or unavailable recommendations.

Question C, 5 marks: Build, subscribe, and monitor

Recommend whether to build internally or subscribe to an external recommendation service. Include fallback and monitoring.

MODEL ANSWER

I recommend a staged hybrid approach: subscribe initially for rapid learning while retaining strategic data, business rules, evaluation, and the option to build core capability.

- Evaluate differentiation, skill, speed, long-term cost, data control, provider quality, and lock-in.
- Maintain an independent evaluation set and product measures instead of trusting provider claims.
- Place the provider behind an internal interface to simplify switching.
- Contract for availability, model changes, price changes, security, and retirement.
- Fallback to repeat purchases, popular available items, or rule-based substitutes. Monitor quality, cost, availability, and complaints.

Recommendation: Revisit internal build when scale, strategic value, and subscription cost justify ownership.

Case Memory Box

Remember

Retail = recommendations must fit the journey and live inventory. Test completed value, not clicks, and protect explicit user choice.

CASE 5: CityServe Multilingual Citizen-Service Assistant

Government, Utilities, and Public Services

Why this case matters

Public services test end-to-end workflow reimagination, multiple stakeholders, language, location uncertainty, accessibility, Agentic authority, privacy, and public-value measures.

Business Case

CityServe receives 80,000 requests each month in six languages. Citizens report street-light failures, waste issues, water interruptions, certificate problems, and property-tax questions by telephone, application, email, and counter visits.

The proposed assistant converts speech to text, identifies language, classifies the issue, links it to a location, checks existing incidents, and recommends the responsible department. Location is missing or uncertain in 22 per cent of requests. Citizens sometimes include personal identifiers and photographs. Management wants the assistant to create field-service work orders automatically.

The product manager must prevent duplicate reports, support citizens who cannot use digital channels, separate confirmed service status from inferred issue type, and decide when an Artificial Intelligence agent may take action.

Direct Topic Links

Topic	Relevance in this case
Workflow redesign	Connect citizen, service desk, department, and field response.
Language	Speech-to-text and Natural Language Processing.
Classification	Issue type, urgency, department.
Location uncertainty	Device location is not always incident location.
Facts and inferences	Known asset and status versus probable issue or location.
Agentic authority	Automatic duplicate linking versus reviewed work-order creation.
Accessibility	Telephone and counter fallback, language support.
Privacy	Personal identifiers, photographs, access, retention.
Metrics	Resolution, duplicate rate, workload, equity, privacy incidents, cost.

How to Read the Case

- Multiple languages and channels
- Location uncertainty
- Public-service equity
- Automatic work orders
- Multiple operating departments

Product Development Link

Citizen voice or text -> language and transcript verification -> account, location, and existing-status retrieval -> issue classification and duplicate check -> clarification or staff review -> work order or information -> field outcome -> citizen status and verified feedback.

Professor-Style Questions and Answers

Question A, 5 marks: Frame the public-service product

Identify the business problem, stakeholders, Artificial Intelligence role, data, and accessibility requirements.

MODEL ANSWER

I will design an assisted request-resolution product, not only a chatbot or automatic ticket generator.

- The city is the buyer; citizens are service users and beneficiaries; service-desk staff, departments, and field teams are operational users.
- Speech-to-text and Natural Language Processing understand requests; classification identifies issue, urgency, and department.
- Retrieve verified account, address, asset, service status, and existing work order before using inference.
- Required data includes channel, language, transcript, selected location, asset, incident, department outcome, and field resolution.
- Maintain telephone and counter alternatives, correction, language access, and privacy controls for identifiers and photographs.

Recommendation: Begin with a unified, accessible request workflow and supervised work-order creation.

Question B, 5 marks: Design facts, inferences, and authority

Design the workflow and explain when the system may act automatically and when a human must review.

MODEL ANSWER

The product will separate verified service information from issue and location inference.

- Fact: verified account, selected address, known asset, published service status, and existing work order.
- Inference: likely issue type or probable location derived from language, photograph, or device location.
- Recommendation: ask a question, merge a duplicate, route to a department, or create a work order.
- Automatic low-risk action may attach a duplicate report to a confirmed incident.
- Missing location, safety issues, sensitive data, expensive work, and conflicting evidence require authorised staff review and audit.

Recommendation: Increase Agentic authority only after a controlled pilot proves dependable outcomes and manageable exceptions.

Question C, 5 marks: Measure and deploy the public product

Propose testing, product, operational, Artificial Intelligence, and public-value measures.

MODEL ANSWER

I will measure whether requests are correctly resolved and citizens receive equitable service, not only conversation volume.

- Test historically, then shadow live classification, then beta with selected departments and languages.
- Artificial Intelligence measures: transcription correction, classification precision and recall, location quality, false duplicate merge.
- Product measures: completed requests, correction, self-service, abandonment, repeat contact, and channel accessibility.

- Operational measures: time to acknowledge and resolve, duplicate ratio, staff workload, automatic versus manual resolution, availability.
- Guardrails: privacy incidents, incorrect work orders, safety misses, and unequal outcomes by area or language.

Recommendation: Expand department by department only after resolution, equity, privacy, and workload guardrails are met.

Case Memory Box

Remember

Public service = redesign the whole service journey. Location, language, accessibility, privacy, and bounded Agentic action matter as much as classification.

CASE 6: LearnPath Student Support and Knowledge Assistant

Education and Learning Technology

Why this case matters

Education tests Generative Artificial Intelligence grounding, hallucination, learner-support prediction, fairness, beneficial intervention, human academic judgment, feedback validation, and provider change.

Business Case

LearnPath University has 18,000 learners and a support team of 35 people. Students ask about registration, fees, examinations, technical access, policies, and learning difficulty. The university wants a Generative Artificial Intelligence assistant trained on policies, course materials, and past support answers.

In evaluation, 78 per cent of answers are fully correct, 14 per cent are partly correct, and 8 per cent contain unsupported information. Policies change several times each term. Management also wants the system to recommend which learners may need additional academic support.

Faculty members are concerned that uncertain policy explanations may be treated as official and that support-risk predictions may unfairly label learners. The external language-model provider can change or retire the model.

Direct Topic Links

Topic	Relevance in this case
Generative Artificial Intelligence	Answer administrative questions in natural language.
Retrieval and grounding	Use approved current sources and show evidence.
Hallucination	Unsupported answer must trigger refusal or escalation.
Prediction	Identify possible need for support, not automatic penalty.
Fairness	Avoid harmful labels and biased features.
Human review	Support staff verify policy; academic adviser reviews learner support.
Feedback	Student reports flag issues; verified staff corrections update knowledge.
Provider risk	Model changes, retirement, evaluation, fallback, migration.
Metrics	Grounded correctness, refusal, correction, support outcome, fairness, workload.

How to Read the Case

- Open-ended generated answer
- Changing policies
- Sensitive learner prediction
- External model dependency
- Human educational judgment

Product Development Link

Student request -> access and policy retrieval -> grounded generation or safe refusal -> support escalation -> verified correction; separately, learner signals -> support-risk inference -> adviser review -> optional beneficial support -> outcome monitoring.

Professor-Style Questions and Answers

Question A, 5 marks: Define the grounded Minimum Viable Product

Recommend the first product scope and prevent unsupported answers from being accepted as official facts.

MODEL ANSWER

I will limit the first version to grounded administrative support using approved, current university sources.

- Cover registration, fees, examination procedures, and technical access where authoritative documents exist.
- Retrieve from approved sources and show policy title, effective date, and supporting link or passage.
- When support is missing or conflicting, state that the answer cannot be confirmed and transfer to staff.
- Do not invent deadlines, grades, exceptions, or academic judgments.
- Student feedback flags a problem, but only verified staff correction updates the approved knowledge base.

Recommendation: Do not launch open-ended answering while eight per cent of outputs remain unsupported without grounding, refusal, and escalation.

Question B, 5 marks: Design responsible learner-support prediction

How should the university identify learners who may need help without creating unfair labels or penalties?

MODEL ANSWER

I will design an offer-of-support product, not an automatic label or restriction.

- The desired outcome is timely support; features may include requested help, missed submissions, and declining engagement subject to purpose and permission.
- Do not use sensitive characteristics merely because they improve prediction; test representation and structural bias.
- Show the output internally as a support signal with understandable evidence, not a public judgment of ability or motivation.
- A qualified adviser reviews the signal and offers optional support; the model does not change grades or enrolment.
- Measure useful support accepted, change after support, false alarms, misses, learner concerns, and group-level performance.

Recommendation: Keep academic judgment and high-impact decisions with qualified humans and use prediction only for beneficial intervention.

Question C, 5 marks: Manage provider change and lifecycle

The external model will be retired in sixty days. Explain migration, testing, communication, fallback, and monitoring.

MODEL ANSWER

I will manage model retirement as a product migration and risk event, not only a technical replacement.

- Inventory functions, prompts, data flows, users, costs, and policies that depend on the model.
- Evaluate replacements using grounded correctness, unsupported output, refusal, language, response time, and cost.
- Use shadow comparison and selected-user beta before live replacement.
- Update support training, monitoring thresholds, documentation, and user communication.

- Retain deterministic search and staff support; pause generative answers if no replacement meets guardrails.

Recommendation: Migrate only when the replacement is at least as safe and useful as the approved baseline.

Case Memory Box

Remember

Education = ground generated answers, use prediction for support rather than punishment, validate feedback, and retain human academic judgment.

CASE 7: MoveFast Delivery Route and Fleet Operations Product

Logistics, Transportation, and Mobility

Why this case matters

Logistics tests real-time recommendations, multiple stakeholders, changing data, regression, ranking, operational override, continuous monitoring, predictive maintenance, and sustainability value.

Business Case

MoveFast operates 1,800 delivery vehicles. Every morning, the planning system creates routes using package addresses, promised delivery windows, vehicle capacity, and expected traffic. During the day, new pickups, traffic incidents, failed deliveries, and customer-priority changes make some routes unsuitable.

Drivers complain that recommendations sometimes direct them to the front entrance even when the loading dock is the correct delivery location. Operations teams want the route system to recalculate continuously. Finance wants lower fuel and overtime cost. Safety staff do not want drivers distracted by frequent route changes.

The company also wants to add electric-vehicle battery-risk prediction. The product manager must connect route recommendation, estimated arrival time, driver correction, operating policy, changing data, and fleet maintenance without giving the system unsafe authority.

Direct Topic Links

Topic	Relevance in this case
Regression	Estimated arrival time, delay, energy use, or battery range.
Recommendation and ranking	Sequence stops and next-best action.
Real-time data	Traffic, pickups, failed stops, priority, charging, vehicle capacity.
Stakeholders	Buyer, dispatcher, driver, recipient, maintenance, safety.
Human override	Driver or dispatcher corrects impractical or unsafe recommendation.
Feedback	Verified delivery point and route outcome improve map and planning.
Monitoring	Recommendation usefulness, delay, miles, corrections, safety, drift.
Predictive maintenance	Battery or vehicle risk linked with route and maintenance planning.
Business value	On-time service, miles, fuel, overtime, fleet availability, sustainability.

How to Read the Case

- Changing live conditions
- Ranked operational actions
- Estimated time

- Driver feedback
- Safety and cost trade-off

Product Development Link

Packages, vehicle, traffic, delivery windows, and map -> route and arrival predictions -> driver or dispatcher review -> delivery outcome and correction -> live recalculation -> operational and business monitoring; battery signals feed a separate maintenance decision workflow.

Professor-Style Questions and Answers

Question A, 5 marks: Frame the logistics product

Identify stakeholders, Artificial Intelligence techniques, data, and the value of the product.

MODEL ANSWER

I will design a dynamic route and fleet-decision product that supports drivers and operations rather than a static morning route plan.

- The company is buyer; dispatchers and drivers are direct users; recipients benefit; maintenance and safety teams are affected stakeholders.
- Regression predicts arrival time, delay, energy use, or range; recommendation ranks route sequence and next action.
- Data includes package address, delivery window, vehicle capacity, traffic, pickups, failed delivery, charging state, and verified delivery point.
- Fixed safety and driving policies remain deterministic; Artificial Intelligence handles changing estimates and ranking.
- Value includes on-time delivery, lower miles, fuel, overtime, and better fleet availability.

Recommendation: Use live recommendation with driver and dispatcher control, not autonomous route changes without operational context.

Question B, 5 marks: Design workflow, correction, and authority

Explain how route changes, uncertainty, user correction, and Human-in-the-Loop should work.

MODEL ANSWER

The system will propose route changes with reason, expected benefit, and effect on later deliveries.

- Confirmed road closure or delivery window is fact; predicted congestion or arrival time is inference.
- A route change is a recommendation and should not distract the driver while driving.
- The driver can confirm loading-dock location or report access failure; dispatcher resolves major priority conflicts.
- High-risk or policy-breaking routes are blocked by rules. Low-confidence changes may be deferred or sent to dispatch.
- Verified delivery-point and outcome corrections update the product through a controlled feedback process.

Recommendation: Keep drivers safe and informed while allowing operations to override recommendations that conflict with real conditions.

Question C, 5 marks: Test, monitor, and extend the product

Recommend an experiment, measures, monitoring, and connection with battery predictive maintenance.

MODEL ANSWER

I will test dynamic routing in selected depots and compare it with the present routing process.

- Use historical simulation, shadow recommendations, then a canary depot rollout.

- Measure on-time delivery, miles, fuel, overtime, driver correction, dispatcher override, failed stops, safety incidents, and customer complaints.
- Monitor performance by depot, route type, weather, vehicle, and time of day because conditions change.
- For electric vehicles, anomaly detection and classification can flag battery risk, but maintenance planners approve inspection and routing restrictions.
- Retain the previous routing method and manual dispatch as fallback; roll back harmful updates.

Recommendation: Scale only when operational value improves without unsafe distraction, unfair workload, or excessive correction.

Case Memory Box

Remember

Logistics = real-time prediction and recommendation must respect driver safety, changing data, operational override, and verified location feedback.

CASE 8: OneSource Enterprise Knowledge and Workflow Assistant

Enterprise Software and Professional Services

Why this case matters

Enterprise software tests retrieval, Generative Artificial Intelligence, confidential-data access, Product Requirements Document, product versus project, source grounding, workflow action, provider risk, productisation, and professional services.

Business Case

OneSource Consulting has 12,000 employees. Project decisions, client methods, policies, technical documents, and meeting notes are stored across email, shared drives, customer systems, and collaboration tools. Employees spend substantial time searching and sometimes use outdated documents.

Management proposes a Generative Artificial Intelligence assistant that answers questions, summarises meetings, extracts action items, and updates project tools. Different employees have different access rights, and some client documents must never be shown outside the project team. The organisation also wants the assistant to create tasks automatically.

A rapid prototype gives useful answers but sometimes combines old and new policies, invents a source, or creates duplicate tasks. Every client uses different tools and terminology. The product manager must define the reusable product core, customer configuration, professional-services role, and safe authority.

Direct Topic Links

Topic	Relevance in this case
Opportunity framing	Reduce search, repeated meetings, lost decisions, and manual logging.
Retrieval and generation	Retrieve authorised current sources before generating an answer.
Access and governance	Respect employee, project, client, and document permissions.
Hallucination and uncertainty	Show source, date, confidence, refusal, and correction.
Minimum Viable Product	One knowledge domain, read-only answer, no autonomous external action.
Agentic authority	Draft task versus automatic task creation and system update.
Product versus project	Reusable core with configurable connectors, terminology, and policy.
Professional services	Integrate systems, map permissions, validate local workflow, train users.
Provider risk	External model, confidentiality, cost, behaviour change, retirement, fallback.

How to Read the Case

- Confidential documents
- Generated answers
- Different permissions
- Automatic workflow action
- High customer-specific variation

Product Development Link

Authorised user question -> identity and permission check -> retrieve current approved sources -> grounded answer with citation or safe refusal -> user correction; meeting input -> draft summary and tasks -> owner approval -> project-system update -> verified outcome and monitoring.

Professor-Style Questions and Answers

Question A, 5 marks: Define the enterprise opportunity and Minimum Viable Product

Identify the business problem, stakeholders, data, Artificial Intelligence role, and first product scope.

MODEL ANSWER

I will build an authorised knowledge-retrieval and answer product before attempting an autonomous enterprise agent.

- The buyer is enterprise management; employees and project teams are users; clients and information owners are affected stakeholders.
- The problem is slow search, outdated information, lost decisions, and repeated manual logging.
- Retrieval finds current approved documents; Generative Artificial Intelligence explains and summarises; Natural Language Processing extracts action items.
- Data includes authorised documents, metadata, version, owner, permissions, meetings, verified project outcomes, and user identity.
- The Minimum Viable Product covers one knowledge domain, read-only grounded answers, citations, safe refusal, and human-reviewed corrections.

Recommendation: Prove dependable retrieval and access control before adding autonomous workflow changes.

Question B, 5 marks: Design trust, access, and Agentic action

Explain grounding, permissions, facts, inferences, generated content, correction, and task-creation authority.

MODEL ANSWER

Every request will begin with identity and permission checks before retrieval or generation.

- A current approved policy passage is fact; a summary is generated content; a proposed action item is a recommendation until the owner confirms it.
- Show source, date, version, and owner. If sources conflict or permission is absent, refuse or escalate.
- Users may report an error, but verified document owners approve corrections.
- The system may create a draft task; the meeting owner approves assignment, due date, and project-system update.
- Log access, source, answer, correction, approval, and action for audit.

Recommendation: Keep generated answers grounded and make enterprise actions reviewable, reversible, and permission aware.

Question C, 5 marks: Productise and source the capability

Explain product versus project, professional services, build-buy-subscribe, monitoring, and provider retirement.

MODEL ANSWER

I will create a reusable core with customer-specific configuration rather than build a new assistant for every client.

- Reusable core: identity, permission-aware retrieval, grounding, citation, answer, correction, task draft, monitoring, and audit.
- Configuration: connectors, terminology, document hierarchy, workflow, roles, and retention policy.
- Professional services integrate systems and validate local process, but the product manager remains accountable for the product promise.
- If subscribing to an external model, control confidential data, maintain an independent evaluation set, monitor cost and behaviour, and isolate the provider for switching.
- Fallback is authorised search and human support. Model retirement requires inventory, replacement testing, user communication, staged migration, and rollback.

Recommendation: Scale through a standard product core, controlled configuration, and an explicit provider exit plan.

Case Memory Box

Remember

Enterprise software = access-controlled retrieval before generation, grounded answers before autonomy, reusable core before customer-specific project work.

Cross-Case Syllabus Map

The same syllabus topics repeat across industries, but the consequence and product design change. Use this map for final recall.

Topic	Healthcare	Banking	Manufacturing	Retail	Public service	Education	Logistics	Enterprise
Classification	Urgency	Fraud	Failure	Substitute	Issue type	Support need	Delay/risk	Intent
Regression	Wait time	Loss estimate	Remaining life	Demand	Resolution time	Completion risk	Arrival time	Cost/time
Anomaly detection	Rare symptom pattern	Suspicious transaction	Sensor abnormality	Unusual demand	Service event	Engagement change	Battery/route event	Access/use anomaly
Recommendation	Department/action	Review action	Inspection priority	Product/substitute	Department/work order	Learning support	Route/stop	Answer/task
Highest-risk error	Missed urgent case	Missed fraud or genuine decline	Missed failure	Unsafe/intrusive recommendation	Wrong work order	Unfair label	Unsafe route	Confidential or false answer
Human reviewer	Clinical staff	Investigator/customer	Planner/technician	Customer/merchandiser	Staff/field team	Adviser/policy owner	Driver/dispatcher	Document/task owner

Universal Three-Question Pattern

Question A

Frame the problem, role, stakeholders, Artificial Intelligence usefulness, technique, and data.

Question B

Design the Minimum Viable Product, workflow, interface, uncertainty, Human-in-the-Loop, and authority.

Question C

Evaluate errors, measures, deployment, monitoring, rollback, sourcing, scale, and final recommendation.

Final One-Page Case Checklist

- Role: Who am I in the case?
- Problem: What outcome must improve?
- People: Who pays, uses, benefits, verifies, and carries risk?
- Artificial Intelligence role: Category, number, anomaly, ranking, language, image, content, or action?
- Data: Source, feature, label, quality, representation, permission, and timing?

- Minimum Viable Product: Which user, workflow, pilot boundary, exclusions, and assumption?
- Trust: Fact, inference, recommendation, unknown, evidence, confidence, correction, escalation?
- Human control: Which person reviews which decision with what authority?
- Proof: False positives, false negatives, model, product, operational, and business measures?
- Operation: Testing, monitoring, fallback, rollback, feedback, provider risk?
- Decision: Launch, pilot, modify, defer, buy, build, subscribe, scale, or stop?

Final memory sentence

Problem -> People -> Artificial Intelligence role -> Data -> Product -> Trust -> Human control -> Proof -> Operation -> Decision.

Source and Scope Note

These eight cases are original learning cases prepared from the professor's case-oriented exam guidance, the course topics reviewed from class recordings and slides through Contact Session 8, the uploaded eBook, and common management-school case archetypes. The electricity and utility style of reasoning remains the base: reframe the complete workflow, maintain the assigned product-manager role, separate facts from inference, name the human reviewer, connect Artificial Intelligence output to business action, and retain evidence, monitoring, fallback, and rollback.

The cases are not predictions of the exact examination industries. They are designed to help transfer the same product-management reasoning to an unfamiliar case.